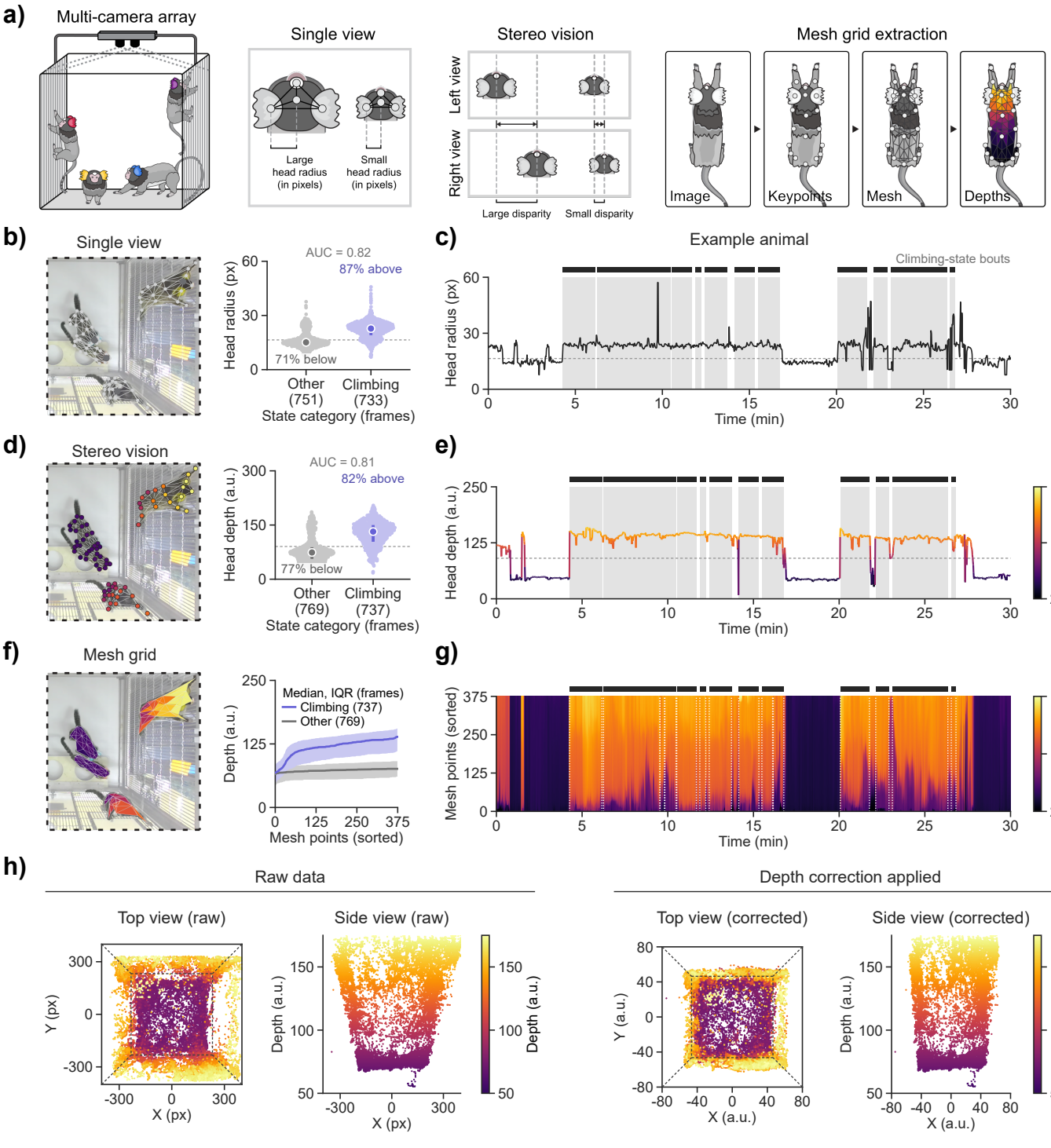

a)
Multi-camera array
Single view
Large head radius (in pixels)
Small head radius (in pixels)
Stereo vision
Left view
Right view
Large disparity
Small disparity
Mesh grid extraction
Image
Keypoints
Mesh
Depths
b)
Single view
AUC = 0.82
87% above
71% below
Head radius (px)
Other (751)
Climbing (733)
State category (frames)
c)
Example animal
Climbing-state bouts
Head radius (px)
Time (min)
d)
Stereo vision
AUC = 0.81
82% above
77% below
Head depth (a.u.)
Other (769)
Climbing (737)
State category (frames)
e)
Head depth (a.u.)
Time (min)
Depth (a.u.)
f)
Mesh grid
Median, IQR (frames)
Climbing (737)
Other (769)
Depth (a.u.)
Mesh points (sorted)
g)
Mesh points (sorted)
Time (min)
Depth (a.u.)
h)
Raw data
Depth correction applied
Top view (raw)
Side view (raw)
Top view (corrected)
Side view (corrected)
Y (px)
X (px)
Depth (a.u.)
Y (a.u.)
X (a.u.)